

Question #1 (10 points): (ILO: A2, B2, B3, Outcome 1)**(Q1.1) (2 points)** Show how the flag register is affected by the addition of A5C2H and 9A3EH**(Q1.2) (3 points)** Indicate whether or not the jump happens in each case. Show your work.

1) ;from the code segment
 CLC
 MOVAH, 55H
 XOR BL, BL
 XOR BL, AH
 MOV CL, 03H
 AND CL, 0FH
 SHR AH, CL
 JNC TARGET

Handwritten notes:
 AH=55
 BL=00 → BL=55
 00000011
 00001111
 00090011
 CL=03
 0101 0101
 0101 0101
 new carry
 Jump not executed

2) ;from the code segment
 CLC
 MOV AL, 2
 SUB AL, 0FFH
 JC TARGET

Handwritten notes:
 AL=00000010
 11111111
 11111111
 00000011
 Jump executed

1.1)
$$\begin{array}{r} 1010\ 0101\ 1100\ 0010 \\ + 1001\ 1010\ 0011\ 1110 \\ \hline 0100\ 0000\ 0000\ 0000 \end{array}$$

CF=1
 SF=0
 ZF=0
 AF=1
 PF=1
 OF=1

(Q1.3) (5 points) Write an assembly program that compares two arrays of integers, if they are identical, the program should print "Matched", otherwise print "Not Matched".

.Model Small

.Stack 64

.Data

X DB 1,2,3,4,5,6

Y DB 1,2,3,4,5,6

W DB 'Matched\$'

Z DB 'Unmatched\$'

.Code

Main Proc Far

Mov Ax, 0data

Mov Es, Ax

Mov Ds, Ax

Mov SI, offset X

Mov DI, offset Y

Mov CL, 06

RePE CMPSB

JE exit

Mov Dx, offset Z

Mov AH, 09

Int 21

J exit2

exit: Mov Dx, offset W

Mov AH, 09

Int 21

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exit 2: Main End P
      End Main
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Question #2 (10 points) (ILO A2, Outcome 1)

(Q2.1) (4 Points) Write a single logical instruction to replace each of the following:

- Clear the odd numbered bits in AX to 0
- Inverts bit number 0,4, 8, 12 in AX
- Set the least significant nibble in AX to 1
- Divide the contents of AX by 2

to replace each of the following:

7 6 5 4 3 2 1 0

0 1 0 1 0 1 0 1

ts

XOR Same $\rightarrow 0$
Different $\rightarrow 1$
(المختلفين)

a) And $Ax, 5555$

b) XOR AX, 1111

121109876543210
10001/0001/0001

c) OR AX, 000F

d) $\text{SHR } \tilde{A}x, 1$

(Q2.2) (6 Points) For the program shown below:

- State the function of this program.
- Determine the memory contents of the 5 consecutive bytes starting from offset input after the execution of this program.
- Determine the CF after execution of line 14.
- Determine the addressing mode of line 17.

1. . DATA
2. Input DB 99H,12H,56H,45H,36H
3. .Code
4. MAIN PROC FAR
5. MOV AX, @DATA
6. MOV DS, AX
7. MOV CH, 04H
8. UP2: MOV CL, 04H
9. MOV SI, offset Input
10. UP1: MOV AL, [SI]
11. MOV BL, [SI+1]
12. CMP AL, BL
13. JNC DOWN
14. MOV DL, [SI+1]
15. MOV AH, [SI]
16. MOV [SI], DL
17. MOV [SI+1], AH
18. DOWN: INC SI
19. DEC CL
20. JNZ UP1
21. DEC CH
22. JNZ UP2
23. MOV AH, 4CH
24. INT 21H
25. MAIN ENDP
26. END MAIN

a) Sort array descending order

b) 99, 56, 45, 36, 12

c) $C_F = 1$

d) Index Relative

Question #3 (10 points) – (ILO: B2, B3, Outcome 2):

(Q3.1) (5 points) Indicate whether each of the following statements is TRUE or FALSE and correct the false ones:-

- F 1- If the crystal oscillator is operating at 30 MHz, the PCLK output of the clock generator is 16 MHz and OSC is 10 MHz. *slow devices* *Min*
- F 2- If the 8087 math coprocessor is not available, the 8086 is operated in a MAX mode.
- F 3- If a CPU has a thirty-two Bit address bus, the maximum amount of memory that can be addressed is 24 MB. *4 G* *$2^{32} \rightarrow 2^{30} \cdot 2^2$* *4 G*
- T 4- The HOLD and HOLDA are required in order to allow the DMA controller to take over control of the bus from the microprocessor.
- T 5- While memory contains both code and data, ports contain data only

$$\begin{aligned} 2^{10} &\rightarrow K \\ 2^{20} &\rightarrow M \\ 2^{30} &\rightarrow G \end{aligned}$$

(Q3.2) (5 points) Trace the following instructions:-

1) MOV DX, 2345H
MOV BX, 4199H
MOV AL, BL
SUB AL, DL
DAA

AL=99
DL=45
AL=54

2) MOV EAX, 4
MOV CL, 7
MUL CL
AAM

3) XOR AH, AH
MOV AL, '6'
ADD AL, '7'
AAA
OR AL, 30H

AH=00

AL=36

AL=6D

AL=60

AL=0D

AL=33

AL=33

AL=26

AL=E9

AL=E9

AL=E9

AL=E9

AL=E9

AL=E9

AL=E9

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AL=E9

AL=E9

AL=E9

AL=6D
-60

+0D
6

13

AH=01

AL=03

AL=33

AH=01

AX=0133

0000 0011
0011 0000
0011/0014

MOV AL, 66H

AND AL, 2FH

OR AL, C3H

NEG AL

XOR AL, 0FH

NOT AL

MOV AL, 12H

CBW

CWD

MOV CL, 03H

DIV CL

0012
3

MOV CL, 03H

DIV CL

MOV CL, 03H

DIV CL

MOV CL, 03H

DIV CL

MOV CL, 03H

DIV CL

MOV CL, 03H

DIV CL

MOV CL, 03H

DIV CL

MOV CL, 03H

DIV CL

MOV CL, 03H

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MOV CL, 03H

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MOV CL, 03H

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MOV CL, 03H

DIV CL

MOV CL, 03H

DIV CL

MOV CL, 03H

DIV CL

MOV CL, 03H

0110 0110
0010 1111

+ 0010 0110
1100 0011

1110 0111

E7

6001 1001
0000 1111

0001 0110

1110 1001

AL=26

AL=26

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Question #4 (10 points): (ILO: B3, Outcome 1)

(Q4.1) (5 points) Answer the following short questions in no more than two lines for each:

- Find the capacity, address, and data pins for the 32K x 8 ROM memory organization.
- A 512K memory chip has 8 pins of data. Find the number of address pins.
- What is the difference between TEST pin and READY pin in 8086?
- State the function of the following signals: BHE and NMI.
- What is the advantage of EEPROM over EPROM?

$$2^5 \cdot 2^{10} = 15 = n = \text{address}$$

data=8

64K x 8

2⁶ · 2¹⁰ = 2¹⁶ → addr

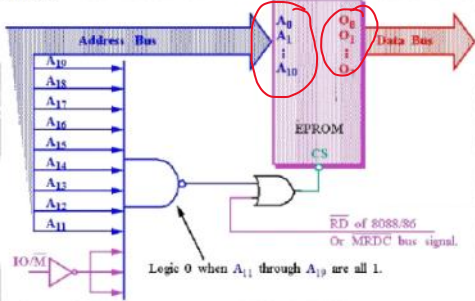
Test → Coordinate coordination between 8088 → 8087 → in max mode
Ready: processor enter a wait state to wait for slower devices

4) BHE → used in 8086 to transfer 16-bit data instead of 8

NMI → Interrupt that can't be ignored

5) EEPROM → erase specific data not all, also can be erased within PC

(Q4.2) (2 points) Find the memory organization and chip capacity for the following EPROM device:

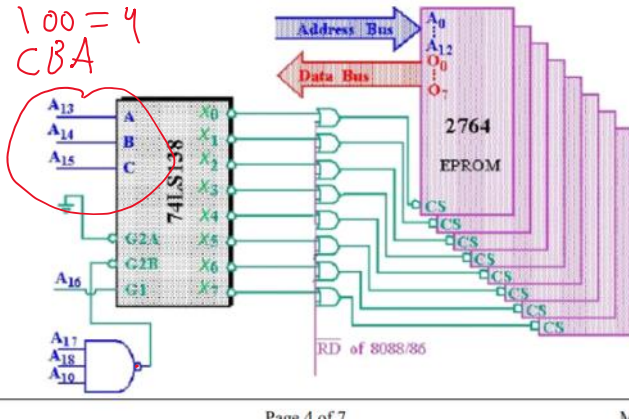


$$2^{11} \rightarrow 2^{10} \cdot 2^1$$

$[2K \times 8] \rightarrow$ organization

$$\text{Capacity} = 16K$$

(Q4.3) (3 points) Find the address range for X_4 and X_7 of the 74LS138 for the following design:



19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1

$F8000 \rightarrow F9FFF$ for X_4

(Q5.2) (5 points) Identify the letter of the choice that best completes the statement or answers the question. Show your work.

- In case of division of doubleword by word, the numerator is in _____ register and the quotient is in _____ register.
a. BX, AX
b. AX, DX
c. DXAX, DX
d. DXAX, AX
DX AX
- If CS = 24F6H, SS = 0801H, SI = 0100H and IP = 634AH, the physical address of the next instruction to be executed is:
a. 83DAEH
b. 2B2AAH
c. 83DA0H
d. None of the above
24F60 + 634A = 2B2AA
- Assume that the current value of CS:IP is 0702:2345 and a jump to address 0802:2323 is executed then the jump is:
a. A NEAR jump
b. A FAR jump
c. An illegal jump
d. None of the above
CS changed so FAR
- If ES = D321H, then the range of physical addresses for the extra segment is:
a. 00000H - 0D321H
b. D3210H - D321FH
c. D3210H - E320FH
d. 0D321H - 1D320H
D3210 + FFFF + 0000
- For the instruction MOV [SI]+12, CL Given CX=7E3DH, SS= 1290H, BP= 3290H, SI= 1234H, DS=23FAH, the physical address of the memory location where the operand is stored:
a. 16DD6
235A

d. 0D321H - 1D320H

5. For the instruction MOV [SI]+12, CL Given CX=7E3DH, SS= 1290H, BP= 3290H, SI= 1234H, DS=23FAH, the physical address of the memory location where the operand is stored:

- 16DD6
- 1A750
- 251E6, 251E7
- 251E6

$$\begin{array}{r} 23FA0 \\ + 1234 \\ + 12 \\ \hline \end{array}$$

Question #6 (10 points) (ILO B2, Outcome 1, Outcome 2)

(Q6.1) (5 points) Use the decoder 74138 to map 32kb of memory starting from address F8000H to FFFFFH into eight 4K x 8 EPROM Chips. Draw the circuit and indicate the start and the end addresses accessible through each EPROM.

$$2^2 \cdot 2^{10} = 2^{12} \rightarrow \text{add}$$

Data = 8

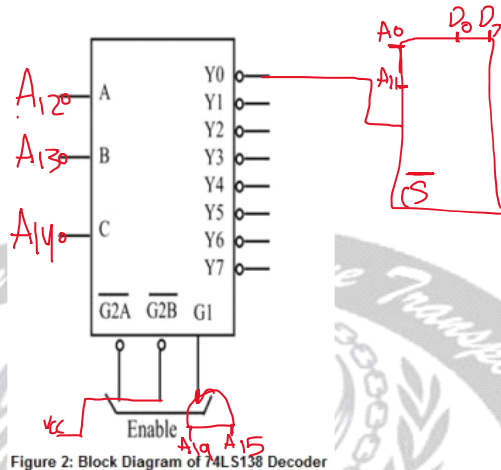
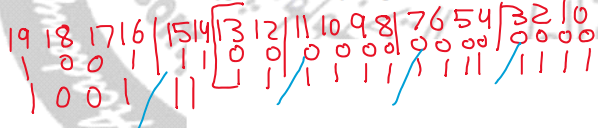


Figure 2: Block Diagram of 74LS138 Decoder

(Q6.2) (3 points) For the design shown below, specify the following:

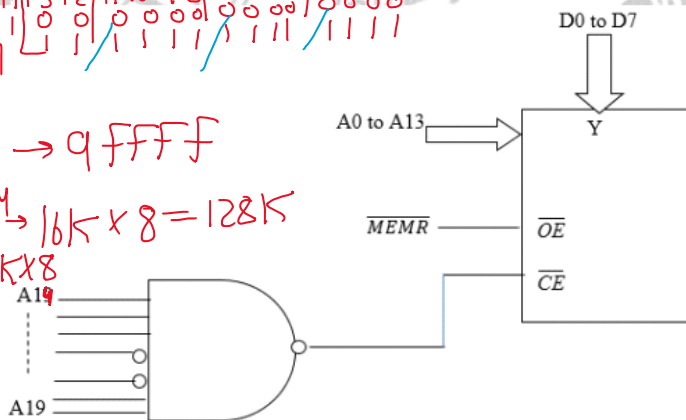
- The address range.
- The memory capacity and organization of the chip.
- The Type of memory chip: RAM or ROM.



$$9C000 \rightarrow 9FFFF$$

b) cap: $2^{14} \rightarrow 16K \times 8 = 128K$
org = $16K \times 8$

c) ROM



(Q6.3) (2 points) Specify NV-RAM chip internal components.